

Added mass

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1 Definition

In hydrodynamics we use the term "added mass" to fully describe a translational motion of the moving body in an ideal liquid. When a body accelerates in the ideal liquid, it causes its environment to move.

To be more clear, the equation of the body's motion will look like:

$$\vec{F} = (m + \Delta m)\vec{a} \quad (1)$$

where m is the mass of the body and Δm is the "added mass".

Or it is the same as saying we have an additional kinetic energy of the liquid in our system which is equal to:

$$\Delta K = \frac{\Delta m v^2}{2} \quad (2)$$

2 Theory of fluid

Incompressibility equation:

For a fluid with constant density, the amount of mass entering the any small volume must equal the amount leaving it.

$$\vec{\nabla} \cdot \vec{u} = 0 \quad (3)$$

If the particles of a liquid do not rotate around their axes, then we consider the flow to be irrotational or potential.

It could be written as:

$$\vec{\nabla} \times \vec{u} = 0 \quad (4)$$

As the rotor of a vector field is zero, this field could be written as a gradient of a scalar potential. Let's call it ϕ . It means the following:

$$\vec{u} = \vec{\nabla} \phi \quad (5)$$

If we put the eq(5) into the eq(3) we get the Laplacian's equation:

$$\nabla^2 \phi = 0 \quad (6)$$

The boundary condition on the surface of the body tells us that the normal component of the fluid's velocity must match the normal component of the body's velocity. It could be written as:

$$\frac{\partial \phi}{\partial n} = v_n \quad (7)$$

where v_n is the projection of the velocity of the body to the normal of the body surface.

3 Deriving the values of "added mass"

Let us consider the translational motion of the ball along the axis x with radius R in a liquid with density ρ

The solution for the Laplace's equation in spherical coordinates with azimuthal symmetry looks like this:

$$\phi(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta) \quad (8)$$

where P_l is a Legendre polynomials which equals:

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} [(x^2 - 1)^l] \quad (9)$$

Or it is the same as:

$$\phi = \left(A_0 + \frac{B_0}{r} \right) P_0 + \left(A_1 r + \frac{B_1}{r^2} \right) P_1 + \left(A_2 r^2 + \frac{B_2}{r^3} \right) P_2 + \dots \quad (10)$$

From eq(9) we get:

$$P_0 = 1; P_1 = x; P_2 = \frac{1}{2}(3x^2 - 1) \dots \quad (11)$$

A_l and B_l are some coefficients which are determined through boundary conditions: We know that the particles of a liquid that are very far away from the body ($r \rightarrow \infty$) does not move ($u \rightarrow 0$). But, from eq(10) we see that as we setting $r \rightarrow \infty$ $\phi \rightarrow \infty$ because of the coefficients $A_1; A_2; A_3 \dots$. So it means:

$$A_1 = A_2 = A_3 = A_l = 0 \quad (12)$$

A_0 is just a formal constant so that we can set it to be zero.

Using eq(7) for our system we get:

$$\frac{\partial \phi}{\partial r}(r = R) = v \cos \theta \quad (13)$$

Taking a partial derivative from eq(10) we could write:

$$\frac{\partial \phi}{\partial r}(r = R) = -\frac{B_0 P_0}{R^2} - \frac{2B_1 P_1}{R^3} - \frac{3B_2 P_2}{R^4} \dots = v \cos \theta \quad (14)$$

Or by using eq(11) we could rewrite it:

$$-\frac{B_0}{R^2} - \frac{2B_1 \cos \theta}{R^3} - \frac{3B_2(3 \cos^2 \theta - 1)}{2R^4} \dots = v \cos \theta \quad (15)$$

By matching terms we get that all the coefficients of B_l are equal to zero except B_1 as it is proportional to the $\cos \theta$:

$$-\frac{2B_1}{R^3} = v; B_1 = -\frac{vR^3}{2} \quad (16)$$

So eq(10) becomes:

$$\phi = -\frac{vR^3 \cos \theta}{2r^2} \quad (17)$$

Now we can find kinetic energy of the fluid:

$$\Delta K = \int \frac{\rho u^2}{2} dV \quad (18)$$

We could rewrite eq(5) as:

$$\vec{u} = \frac{\partial \phi}{\partial r} \hat{r} + \frac{\partial \phi}{\partial \theta} \frac{1}{r} \hat{\theta} \quad (19)$$

We could find the components of u :

$$u_r = \frac{\partial \phi}{\partial r} = \frac{vR^3 \cos \theta}{r^3} \quad (20)$$

$$u_\theta = \frac{vR^3}{2r^3} \sin \theta \quad (21)$$

And adding those components leaves us with:

$$\Delta K = \int \frac{\rho(u_r^2 + u_\theta^2)}{2} dV \quad (22)$$

But for a very tiny volume of a thin ring we could write:

$$dV = dS dr = 2\pi r^2 \sin \theta d\theta dr \quad (23)$$

Using eq(23) we could rewrite formula for ΔK :

$$\Delta K = \frac{2\pi \rho v^2 R^6}{2} \int_R^\infty \int_0^\pi \frac{1}{r^6} (\cos^2 \theta + \frac{\sin^2 \theta}{4}) r^2 dr \sin \theta d\theta = \pi \rho v^2 R^6 \int_R^\infty \frac{dr}{r^4} \int_0^\pi (\cos^2 \theta + \frac{\sin^2 \theta}{4}) \sin \theta d\theta \quad (24)$$

By integrating and putting in limits of our integral gives us:

$$\Delta K = \frac{\rho \pi R^3 v^2}{3} = \frac{2\rho \pi R^3}{3} \frac{v^2}{2} \quad (25)$$

So we get our value for "added mass" for a ball with radius R in the ideal liquid with density ρ :

$$\boxed{\Delta m = \frac{2\rho \pi R^3}{3}} \quad (26)$$